

Improper Integrals

There are two types of improper integrals, Type I and Type II.

Type I:

i. $\int_a^{\infty} f(x)dx =$

ii. $\int_{-\infty}^b f(x)dx =$

iii. $\int_{-\infty}^{\infty} f(x)dx =$

$\int_a^{\infty} f(x)dx$ and $\int_{-\infty}^b f(x)dx$ are _____ if the

limit exists, and _____ if the limit does not exist.

Examples – Type I

Example 1:

$$\int_0^{\infty} \frac{3x^2}{\sqrt{1+x^3}} dx$$

Example 2:

$$\int_{-\infty}^0 x e^{2x} dx$$

Example 3:

$$\int_{-\infty}^{\infty} x^3 e^{-x^4} dx$$

Type II:

- i. $f(x)$ is continuous on $[a, b)$ and discontinuous at b , then

$$\int_a^b f(x)dx =$$

- ii. $f(x)$ is continuous on $(a, b]$ and discontinuous at a , then

$$\int_a^b f(x)dx =$$

- iii. If $f(x)$ has a discontinuity at c , where $a < c < b$, then

$$\int_a^b f(x)dx =$$

Examples – Type II**Example 1:**

$$\int_0^1 \frac{3}{x^5} dx$$

Example 2:

$$\int_2^3 \frac{1}{\sqrt{3-x}} dx$$

Example 3:

$$\int_0^5 \frac{x}{x-2} dx$$